Brain Flow

One developer. Full team. AI orchestration.

A portfolio case study by Mirko · 2026

THE PREMISE

What if one person could do it all?

BrainFlow was built to answer this: a full product — designed, engineered, documented, and pitched — by a single developer directing specialized AI agents.

- Design
- Engineering
- Documentation
- Presentation
- Architecture Audit

AI Agent Roles

Five specialized agents — each with a defined scope, skill prompt, and execution context

Explorer

Researches the codebase, finds patterns, deep-dives into unfamiliar territory

Designer

Visual identity, color decisions, layout, slides — enforces design tokens

Builder

Components and canvas node types built from shared primitives

Auditor

Architecture review, performance, accessibility, code quality checks

Documenter

Records decisions, rationale, progress — builds the case history

The Workflow

How a single prompt becomes a production-ready output

01 Idea

Mirko defines the goal: a feature, design decision, or document to create

02 Skill

The right skill is invoked — /design, /component, /node, /document, /audit

03 Agent

Claude reads the skill prompt + full codebase context, executes autonomously

04 Review

Mirko reviews, gives feedback, iterates — humans stay in creative and strategic control

Tech Stack

Production-grade tools — modern patterns, full type safety, component-level documentation

Next.js 16

FRAMEWORK

App Router with breaking changes navigated from scratch

React 19

UI LIBRARY

Server Components + concurrent features

TypeScript 5

LANGUAGE

Strict typing across the entire codebase

ReactFlow v12

CANVAS

React Flow v12 — the node graph engine

Claude API

AI ENGINE

Run Flow synthesis + Digital Twins intelligence

Tailwind CSS v4

STYLING

Design tokens, glassmorphism, dark mode utilities

Storybook 10

DOCS

Component documentation and visual testing

Vitest + Playwright

TESTING

Unit, integration, and end-to-end coverage

Architecture Decisions

Three foundational choices that shaped every component, every node, every pixel

UI ARCHITECTURE

shadcn-first Components

Check shadcn/ui and the existing component library before writing any custom UI. This creates system-wide consistency and prevents reinventing primitives. Every node type is composed from shared pieces: NodeHandles, NodeActions, NodeHeader, EditableField.

DESIGN SYSTEM

Glassmorphism Design System

All app overlays (header, toolbar, status bar, prompt bar) use GLASS_CONTAINER_CLASS with backdrop-blur. Canvas nodes are exempt — they use solid --node-bg for legibility. The whole system is token-driven: --glass-bg, --radius-*, --node-bg in globals.css.

EXTENSIBILITY

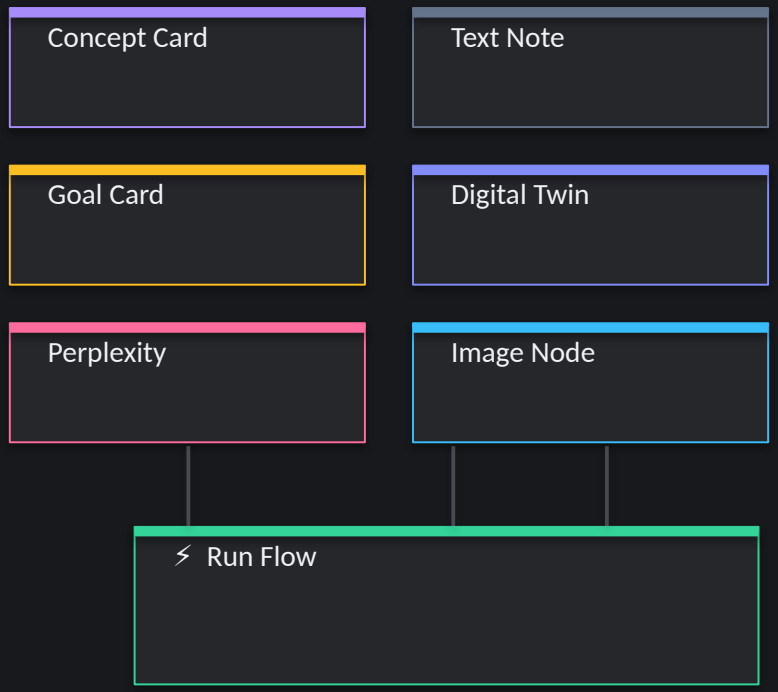
Node Composition Pattern

Every canvas node shares the same anatomical pattern. Adding a new node type is predictable: implement NodeHandles + NodeActions + NodeHeader + EditableField, apply the accent color from the design token map, register in the node factory.

BrainFlow

The Product

A canvas-based brainstorming tool where ideas become structured AI outputs. Build a visual graph — connect concepts, goals, questions, and Digital Twin personas — then synthesize everything through a single Run Flow node.



The Core Loop

1. Create nodes on the canvas
2. Connect them freely
3. Attach to a Run Flow node
4. Get structured AI synthesis

Digital Twins

AI personas with programmable behavior — a simulated team living on your canvas

Contraddici

Devil's advocate — challenges every assumption, pushes back on ideas

Collabora

Builds on ideas, suggests expansions and new connections

Analizza

Critical breakdown — finds gaps, blind spots, logical inconsistencies

Provoca

Pushes thinking to the extreme — radical reframes and bold provocations

What Was Built

One developer, directing AI across design, engineering, documentation, and presentation

7+

Node Types

Text, Concept, Image, Run Flow, Goal, Perplexity, Digital Twin — each composable

5

AI Agents

Explorer, Designer, Builder, Auditor, Documenter — each with defined skill prompts

100%

Design System

Token-driven, glassmorphism, dark mode, consistent across every component

1

Developer

Full orchestration of design + engineering + docs + pitch by a single human

One person.
Full orchestration.

Not replacing a team — proving what's possible
when human creativity directs AI execution.

BrainFlow · Mirko · 2026